Investigating Max Pressure Jump in Mixed Phase

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In this study, we investigate the first-order phase transition between hadronic and quark matter in the core of neutron stars by using a quadratic interpolation method. Our approach modifies the Maxwell construction to include an additional pressure term that accounts for structural effects in the mixed phase, often referred to as the "pasta phase."[1] By implementing a piecewise polytropic equation of state (EOS) for the hadronic and quark phases, we create a smooth transition across the mixed phase[2]. This model enables us to examine the maximum achievable pressure jump while maintaining continuity and stability between the phases. We explore how varying parameters, specifically the polytropic index Γ and critical chemical potential $\mu_{\rm crit}$, affect the phase transition properties. This approach allows us to see how the mixed phase changes with respect to different EOS.

I. INTRODUCTION

The Hadron-Quark, a first-order phase transition, occurs in the core of highly interacting compact stars. We substitute the Maxwell construction between a low-density hadron and a high-density quark matter phase, also known as the pasta phase, without explicitly computing geometrical structures. We do this by using the quadratic interpolation method for the hadronic and quark matter transition phases. This model incorporates an additional pressure term to account for the structural effects in the mixed phase and their influence on the critical pressure within the Maxwell construction Ayriyan and Grigorian [2].

We are well aware of the properties of the equation of state around the nuclear saturation density, n_0. By setting the values of μ and P at this saturation density, we want an equation that gives us the minimum possible value for the additional pressure term for a given P and μ from which we can have the maximum smoothness an EOS can have when we are given two EOS that have a first-order phase transition. If we can find an expression for the additional pressure term, we can apply it to any given equation of state (EOS).

II. MOTIVATION

Understanding the structure and stability of hybrid compact stars through the investigation of their mixed phases is crucial. Therefore, the development of a mathematical treatment to study these phase transitions is necessary. However, the lack of well-studied microphysics of these transitions limits our understanding. Addressing these gaps is crucial for accurately describing matter in such extreme environments and connecting theory with astrophysical observations.

III. FORMALISM

First, we use piecewise polytrope to build EOS for the hadronic and quark phases. Next, we search for a quadratic interpolation curve that can replace the mixed phase, exhibits the highest smoothness, and traverses various critical pressure values, specifically the intersection pressure of the hadronic and quark polytropes.

A. PIECEWISE POLYTROPE EOS WITH A FIRST- ORDER PHASE TRANSITION

To investigate the possibility of describing compact star phenomena using a piecewise polytropic EOS at supersaturation densities, we define one of the polytropes as a constant pressure region with $P = P_{\rm crit}$, where a jump in energy density ϵ due to a first-order phase transition fulfills the Seidov constraint [3].

$$\frac{\Delta\epsilon}{\epsilon_{\rm crit}} \geq \frac{1}{2} + \frac{3}{2}\frac{P_{\rm crit}}{\epsilon_{\rm crit}}, \tag{1}$$

The supersaturation density region is divided into three polytropic regions where the n_1 just above the nuclear saturation density ($n_o = 0.15\,\mathrm{fm}^{-3}$) and polytrope equation fulfill

$$p_i(n) = k_i(n)^{\Gamma_i} \tag{2}$$

$$\begin{aligned} &i = 1 : n_1 \geq n \geq n_{12} \\ &i = 2 : n_{12} \geq n \geq n_{23} \\ &i = 3 : n \geq n_{23} \end{aligned} \tag{3}$$

For the above regions at zero temprature we use the known thermodynamical relations for maxwell construction of a first order phase transition [4],

$$P(n) = n^2 \frac{d\left(\frac{\varepsilon(n)}{n}\right)}{dn}, \tag{4}$$

$$\frac{\varepsilon(n)}{n} = \int dn \frac{P(n)}{n^2} = \int dn\, \kappa n^{\Gamma-2} = \frac{\kappa n^{\Gamma-1}}{\Gamma-1} + C, \quad (5)$$

Here the integration constant C will be m_0, from the condition $\varepsilon = m_o n$ at $n \to 0$

$$\mu(n) = \frac{P(n)+\varepsilon(n)}{n} = \frac{\kappa\Gamma}{\Gamma-1} n^{\Gamma-1} + m_0, \quad (6)$$

$$n(\mu) = \left[(\mu - m_0)\frac{\Gamma-1}{\kappa\Gamma}\right]^{\frac{1}{\Gamma-1}}, \quad (7)$$

$$P(\mu) = \kappa\left[(\mu - m_0)\frac{\Gamma-1}{\kappa\Gamma}\right]^{\frac{\Gamma}{\Gamma-1}}. \quad (8)$$

B. EQUATION OF STATE

From the above relations we construct EOS with a phase transition.we have divided i = 3 into two branches i = 3a and i = 3b their respective density range will be $n_{23} \geq n \geq n_{34}$ and $n \geq n_{34}$ where $n_1 = 0.165 fm^{-3}, n_{12} = 0.32 fm^{-3}, n_{23} = 0.5344 fm^{-3}, n_{34} = 0.7500 fm^{-3}$

TABLE I. Values of Γ_3, κ_3, and $m_{0,3}$ for different parameter sets

	Γ_3	κ_3 [MeV fm$^{3(\Gamma_3-1)}$]	$m_{0,3}$ [MeV]
i = 1	4.921	17906.60	939.56
i = 2	0	63.177	1201.75
i = 3a	4.00	774.375	1031.815
i = 3b	2.80	548.309	958.553

The EOS has following set of properties :

$$P_{\text{crit}} = 63.177\,\text{MeV fm}^{-3},$$

$$\varepsilon_{\text{crit}} = 318.26\,\text{MeV fm}^{-3},$$

$$\Delta\varepsilon = 253.89\,\text{MeV fm}^{-3}.$$

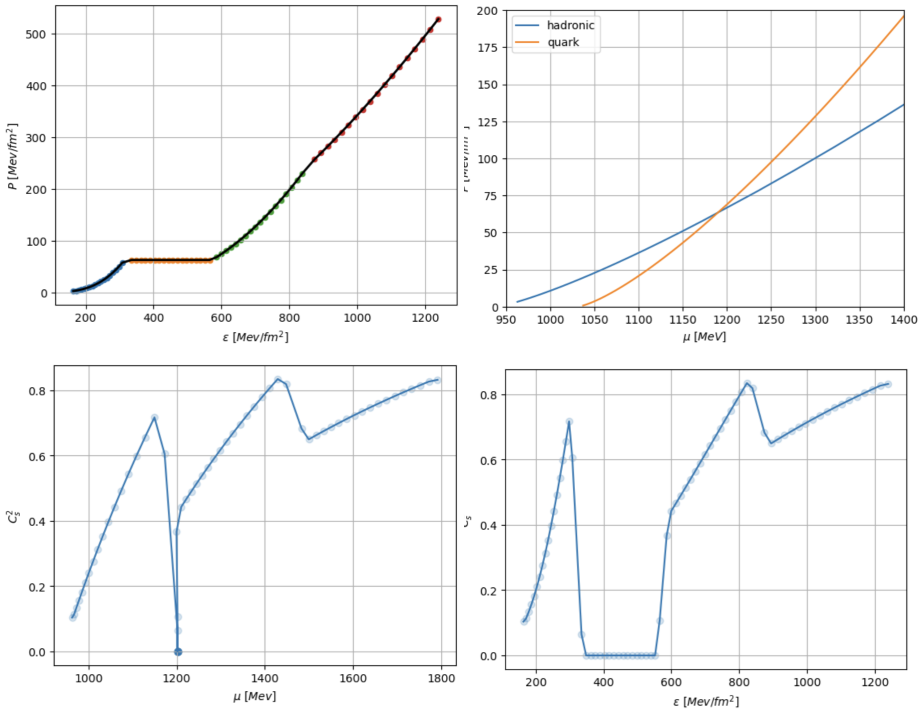


FIG. 1. combined EOS

C. QUADRATIC INTERPOLATION FOR THE FIRST- ORDER PHASE TRANSITION

Let $P_H(\mu), P_Q(\mu)$ be EOS describe the hadronic and quark matter dependence of pressure on chemical potential respectively.P_C be the critical pressure where both the hadronic and quark EOS have same chemical potential , the equation is given by

$$P_H(\mu_c) = P_Q(\mu_c) = P_C \tag{4}$$

We can modify the maxwell construction on the assumption of both the EOS are changing due to their interactions. The mixed phase EOS is given by [2].

$$P_M(\mu) = \alpha(\mu - \mu_c)^p + \beta(\mu - \mu_c)^q + P_c + \Delta P \tag{5}$$

We consider quadratic form (p=2,q=1) of the equation ,

$$P_M(\mu) = \alpha(\mu - \mu_c)^2 + \beta(\mu - \mu_c)^1 + P_c + \Delta P \tag{6}$$

Here we are constructing a quadratic equation(Mixed Phase) around the point $P + \Delta P$ such that the jump from hadronic Phase to mixed phase - quark phase is smooth so there will be $\mu_{cH} and \mu_{cQ}$ where the change from hadronic phase to mixed phase and mixed phase to quark phase happens respectively so there are four unknowns along with α, β [2].Therfore we obtain alpha and beta as

$$\begin{cases} \alpha = \frac{1}{\mu_{cQ}-\mu_{cH}} \left(\frac{P_Q(\mu_{cQ})-(\Delta P+P_c)}{\mu_{cQ}-\mu_c} - \frac{P_H(\mu_{cH})-(\Delta P+P_c)}{\mu_{cH}-\mu_c} \right) \\ \beta = \frac{P_H(\mu_{cH})-(\Delta P+P_c)}{\mu_{cH}-\mu_c} + \frac{P_Q(\mu_{cQ})-(\Delta P+P_c)}{\mu_{cQ}-\mu_c} - \frac{P_Q(\mu_{cQ})-P_H(\mu_{cH})}{\mu_{cQ}-\mu_{cH}}, \end{cases} \tag{7}$$

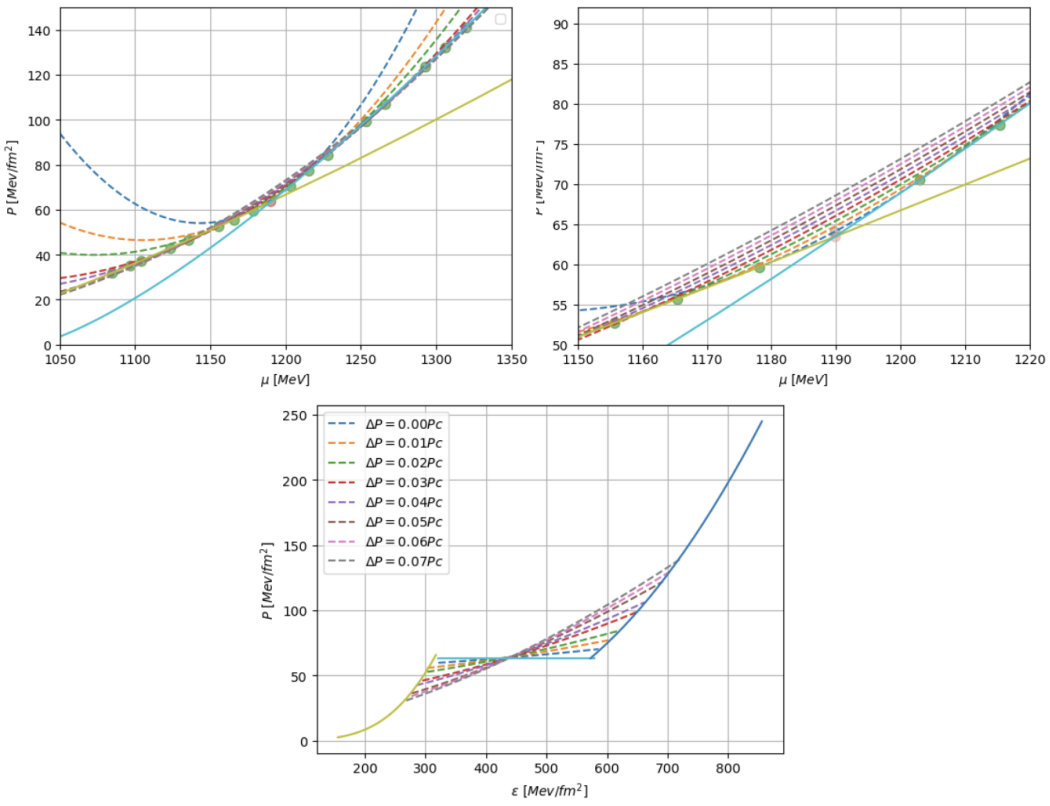


FIG. 2. The quadratic interpolation for the additional pressure of the constructed Piecewise polytrope EOS in fig1.

D. MAXIMUM INTERPOLATION ACHIEVABLE

In equation 7, if we fix the μ_{cH} value to the initial value of the hadronic phase, we desire an μ_{cQ} such that the mixed phase EOS passes through both of the points and it should also pass through any of the additional pressure points, i.e., $P_c + \Delta P$ ($\Delta P \in [0.1, 0.9]P$). The

mixed phase is subject to the following total constraints:

Equality Constraint

The mixed phase EOS $P_M(\mu)$ should always pass through the initial chemical potential and pressure point of hadronic phase which will be our μ_{cH} and P_{cH}.

Inequality Constraint

The mixed phase EOS $P_M(\mu)$ curve should always stay above the hadronic and quark phase or it can intersect.

Optimization Technique

We require the values of α and β of the quadratic equation $P_M(\mu)$ such that this equation satisfy all the necessary constraints and the curve($P_M(\mu)$) should intersect the $P_Q(\mu)$.So we develop a optimization technique where the quadratic curve around the point $P_c + \Delta P$ for all the specified range of ΔP and the algorithm Equality Constraints always ensures the quadratic function passes through ($P_{cH}(P_{min})$,$\mu_{cH}(\mu_{min})$), creating continuity with the hadronic phase.Inequality Constraints Ensures the quadratic function stays above specified points in the hadronic and quark matter phases. These constraints reflect physical limitations that the model must respect to maintain physical feasibility in both phases.

The Model will consider these as its inputs and it will predict the optimal values of α and β along with the respective optimal pressure (P_c+ΔP) values and optimal quadratic fit($P_M(\mu)$ [5].

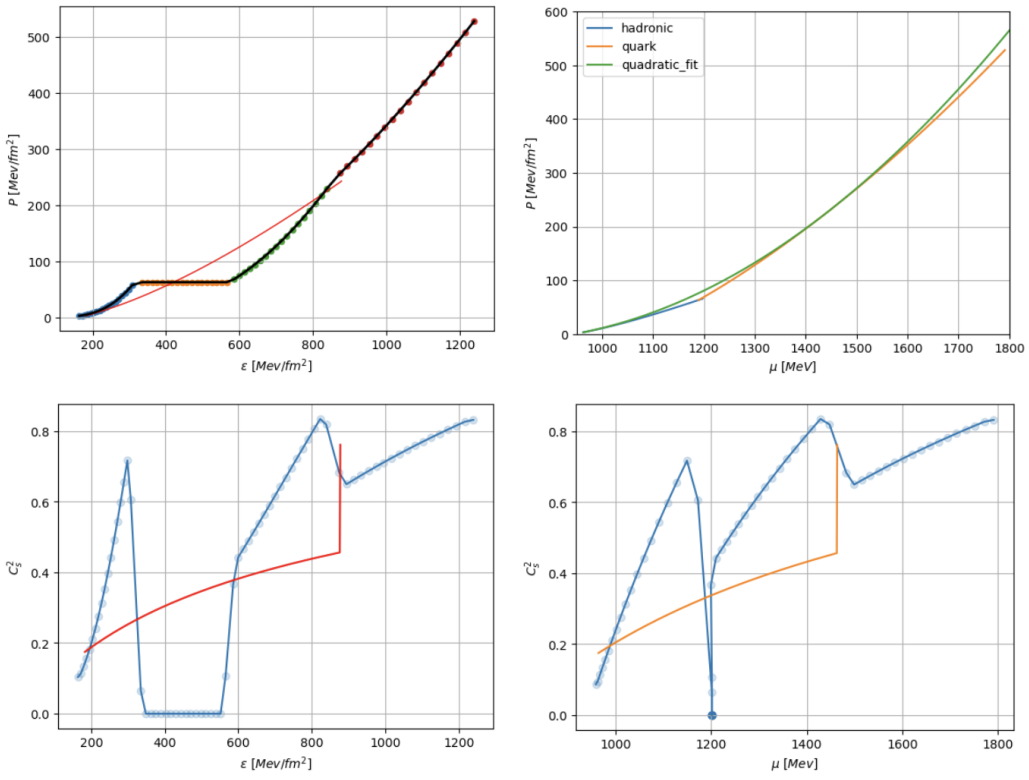


FIG. 3. Optimal Quadratic fit for given EOS

IV. RESULTS

1. *Relation of optimal pressure to Different Γ_Q.*

By modifying the values of Γ of Quark in the equation of state (EOS), we generate an alternative quadratic relationship, which offers an opportunity to investigate how different gamma quark values impact the characteristics of the phase transition. Specifically, adjusting Γ_Q influences the stiffness and overall shape of the EOS, thereby altering the predicted behavior of mixed phase pressure as a function of μ. This model deepens our understanding of the mixed phase that governs the transition between hadronic and quark phases in neutron star matter.

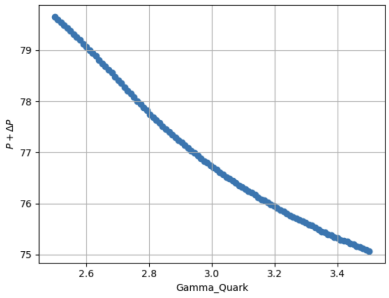


FIG. 4. Relation Between Γ_Q and $P_c + \Delta P$

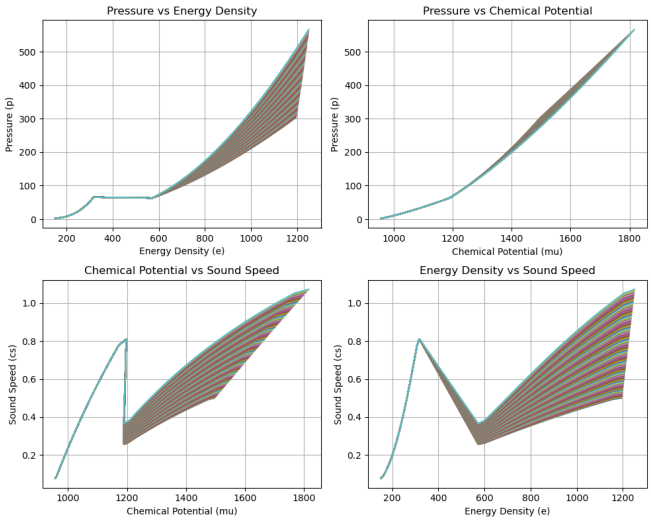


FIG. 5. EOS with different Γ_Q

2. *Relation of optimal pressure to Different $\mu_{critical}$.*

Now we change the point where phase transition occurs i.e, for same pressure(P_{crit}) but different μ_{crit} .

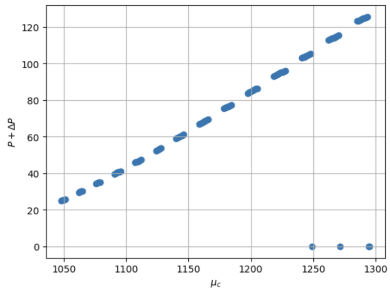


FIG. 6. Relation between different μ_{crit} and $P_c + \Delta P$

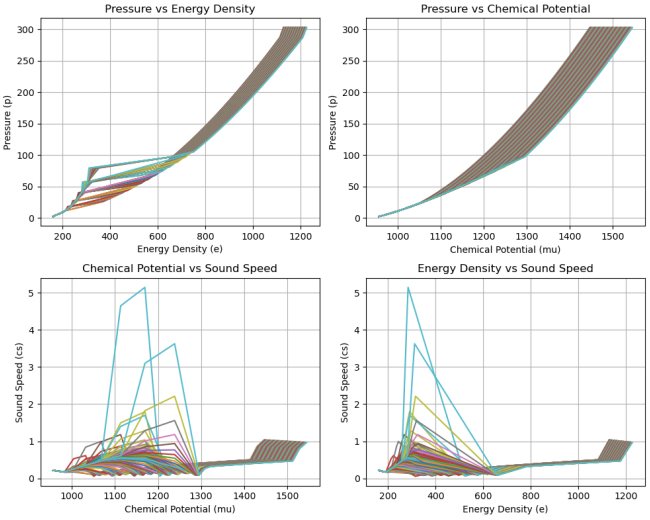


FIG. 7. EOS with different μ_{crit}

V. FUTURE WORK

Understanding the Physical interpolation of EOS with respect to Neutron Stars.

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